Software Requirements Specification

for

ROS2-Based Autonomous Warehouse Fleet Management System

Implementing a Dynamic Lane Reservation Strategy for Multi-Robot Traffic Coordination

Version 1.0

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Revision History

Name	Date	Reason For Changes	Version
Diya Jabin	18 July 2026	Initial creation of the Software Requirements Specification	1.0

1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to define the functional and non-functional requirements for the ***Autonomous Warehouse Fleet Management System (AWFMS)****. The system aims to coordinate multiple autonomous mobile robots operating within a warehouse environment by managing delivery tasks, allocating robots, monitoring robot status, coordinating warehouse traffic, preventing collisions, and handling dynamic obstacle scenarios through a centralized fleet management and traffic coordination system. This document serves as the primary reference for developers, testers, and project evaluators throughout the software development lifecycle*

1.2 Document Conventions

- *IEEE 830 Software Requirements Specification standard is followed.*
- *Functional requirements are labelled* ***FR-x****.*
- *Non-functional requirements are labelled* ***NFR-x****.*
- *High priority requirements are essential for project completion.*
- *Medium priority requirements improve usability.*
- *Low priority requirements are considered future enhancements*

1.3 Intended Audience and Reading Suggestions

This Software Requirements Specification (SRS) is intended for software developers, test engineers, project supervisors, warehouse operators (end users), and future developers involved in the design, implementation, testing, maintenance, and evaluation of the Autonomous Warehouse Fleet Management System.

The document is organized into six major sections. Section 1 introduces the project objectives, scope, and references. Section 2 provides an overview of the proposed system, including its architecture, operating environment, user classes, and design constraints. Section 3 describes the external interfaces of the system, including user, hardware, software, and communication interfaces. Section 4 specifies the functional requirements and system features in detail. Section 5 defines the non-functional requirements, while the appendices contain the glossary, analysis models, and future enhancements.

Readers are encouraged to begin with the Introduction and Overall Description sections before proceeding to the detailed functional and non-functional requirements. Developers and future maintainers should read the document sequentially, while testers may primarily refer to the System Features and Non-Functional Requirements sections.

1.4 Product Scope

The proposed system simulates an autonomous indoor warehouse where multiple autonomous robots transport packages between predefined locations such as storage shelves, loading stations, and charging stations.

A centralized Fleet Management System will:

- *Register available robots*
- *Accept delivery requests*
- *Allocate delivery tasks*
- *Monitor robot status*
- *Coordinate warehouse traffic*

- *Prevent collisions*
- *Handle dynamic obstacle scenarios*
- *Maintain delivery records*
- *Provide real-time warehouse information through an interactive dashboard*

The project primarily focuses on centralized fleet coordination, traffic management, and scalable software architecture rather than physical robot hardware implementation .The modular architecture has been designed to support future integration with physical robots, cloud-based monitoring systems, AI-driven task scheduling, computer vision, and IoT-enabled warehouse automation without significant architectural modifications.

1.5 References

1. *IEEE Computer Society, IEEE Recommended Practice for Software Requirements Specifications (IEEE Std. 830-1998).*
2. *Open Robotics, ROS 2 Documentation.*
3. *Open Robotics, Gazebo Simulator Documentation.*
4. *Navigation2 (Nav2) Documentation.*
5. *Python Software Foundation, Python 3 Documentation.*
6. *Streamlit Documentation*

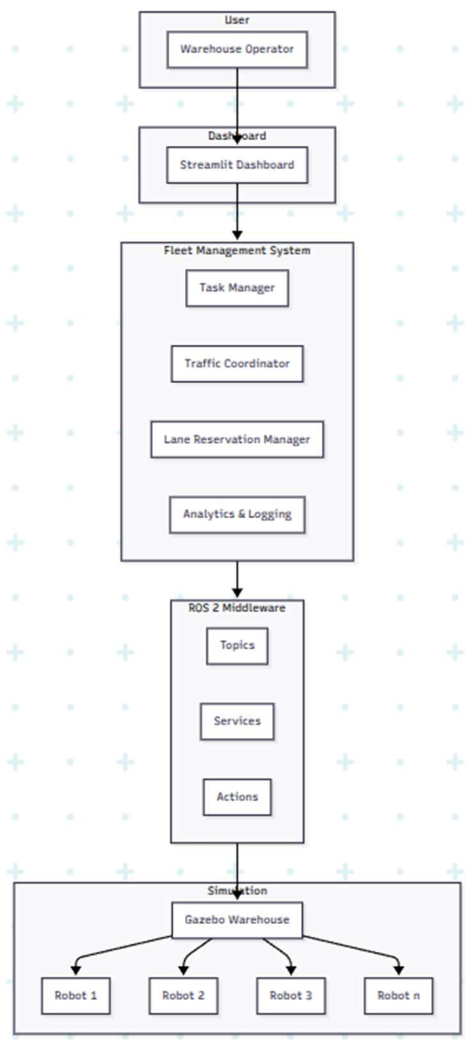
2. Overall Description

2.1 Product Perspective

*The **Autonomous Warehouse Fleet Management System (AWFMS)** is a standalone software system designed to simulate and coordinate multiple autonomous mobile robots operating within an indoor warehouse environment. The system is developed using **ROS 2 Jazzy** and **Gazebo**, providing a realistic simulation platform for evaluating fleet coordination strategies before deployment on physical robots.*

The AWFMS follows a centralized fleet management architecture in which a Fleet Management System acts as the decision-making unit. The Fleet Manager communicates with multiple autonomous robots, assigns delivery tasks, coordinates warehouse traffic, manages dynamic lane reservations, and monitors the operational status of each robot. A Streamlit-based dashboard provides warehouse operators with real-time visualization of robot locations, task progress, lane occupancy, and system analytics.

The modular architecture enables future integration with physical robots, cloud-based warehouse monitoring, AI-driven task scheduling, computer vision, and IoT-enabled warehouse automation without requiring significant architectural modifications



2.2 Product Functions

The Autonomous Warehouse Fleet Management System provides centralized management and coordination of multiple autonomous warehouse robots. The primary functions of the system include:

- *Robot registration and fleet initialization.*
- *Creation and management of delivery requests.*
- *Intelligent task allocation to available robots.*
- *Real-time monitoring of robot status and locations.*
- *Dynamic lane reservation and traffic coordination.*
- *Collision prevention through centralized lane management.*
- *Obstacle detection and route reassignment.*
- *Delivery status tracking and event logging.*
- *Live warehouse visualization through an interactive dashboard.*
- *Generation of warehouse performance analytics and reports*

2.3 User Classes and Characteristics

The Autonomous Warehouse Fleet Management System is intended to be used by different categories of users, each with specific roles and responsibilities.

Warehouse Operator

The warehouse operator is responsible for creating delivery requests, monitoring robot activities, viewing warehouse status, receiving system alerts, and analyzing delivery performance through the dashboard. This user interacts with the system frequently during normal warehouse operations.

Fleet Administrator

The fleet administrator supervises the overall warehouse fleet, configures system settings, monitors fleet performance, reviews operational analytics, and ensures efficient traffic coordination among robots.

Autonomous Robot

Each autonomous robot acts as an independent system entity capable of receiving delivery tasks, navigating the warehouse, detecting obstacles, requesting lane reservations, and periodically reporting its operational status to the Fleet Management System.

Future Developers

Future developers are expected to extend the system by integrating advanced technologies such as artificial intelligence, cloud computing, IoT devices, computer vision, or physical robots while preserving the modular software architecture

2.4 Operating Environment

The Autonomous Warehouse Fleet Management System shall operate in the following software environment:

- *Operating System: Ubuntu 24.04 LTS*
- *Middleware: ROS 2 Jazzy*
- *Simulation Platform: Gazebo Harmonic*
- *Programming Language: Python 3*
- *Dashboard Framework: Streamlit*
- *Database: SQLite*
- *Visualization Tool: RViz2*
- *Version Control: Git and GitHub*

The system is intended to run on a standard personal computer capable of supporting ROS 2 simulation and visualization tools

2.5 Design and Implementation Constraints

The development of the Autonomous Warehouse Fleet Management System is subject to the following constraints:

- *The system shall be developed using the ROS 2 Jazzy middleware.*
- *The warehouse environment shall be simulated using Gazebo Harmonic.*
- *A centralized fleet management architecture shall be implemented.*
- *Communication between software components shall utilize ROS 2 topics, services, and actions.*
- *The warehouse layout shall remain predefined during simulation.*
- *Dynamic lane reservation shall be implemented within the simulated warehouse environment.*
- *The dashboard shall be developed using Streamlit.*
- *The software shall follow a modular architecture to facilitate future scalability and integration with additional technologies.*

2.6 User Documentation

The project shall provide the following documentation to assist users and future developers:

- Software Requirements Specification (SRS)
- User Manual
- Installation Guide
- System Architecture Documentation
- ROS 2 Package Documentation
- GitHub Repository Documentation (README)
- API Documentation (if applicable

2.7 Assumptions and Dependencies

The successful operation of the Autonomous Warehouse Fleet Management System is based on the following assumptions and dependencies:

- *The warehouse environment remains predefined and static throughout the simulation.*
- *All autonomous robots are capable of localization within the simulated environment.*
- *Reliable communication exists between the Fleet Management System and all robots.*
- *The ROS 2 Navigation Stack (Nav2) is properly configured and operational.*
- *Robot movement is restricted to predefined warehouse pathways.*
- *All robots are registered with the Fleet Management System before task allocation begins.*
- *The simulation is executed on hardware capable of supporting ROS 2 and Gazebo without significant performance degradation.*

3. External Interface Requirements

3.1 User Interfaces

The primary user interface of the Autonomous Warehouse Fleet Management System is a ***Streamlit-based web dashboard*** *designed for warehouse operators and system administrators. The dashboard provides an intuitive interface for monitoring and managing fleet operations.*

The dashboard shall provide the following functionalities:

- *Register and monitor autonomous robots.*
- *Create and assign delivery tasks.*
- *Display real-time robot locations within the warehouse.*
- *Display lane occupancy and reservation status.*
- *Notify users of obstacle detection and traffic conflicts.*
- *Display delivery progress and task completion status.*
- *Generate warehouse performance analytics and operational reports.*

The interface shall be designed to be simple, responsive, and easy to operate without requiring extensive technical knowledge

3.2 Hardware Interfaces

Although the current implementation is simulation-based, the software architecture has been designed to support future integration with physical autonomous robots.

The proposed hardware interface includes:

- *ESP32-based autonomous mobile robots.*
- *Ultrasonic sensors for obstacle detection.*
- *IR sensors for line following.*
- *DC motors controlled using an L298N/L293D motor driver.*
- *Battery monitoring module (future enhancement).*
- *Wi-Fi module for wireless communication.*

The simulation environment developed in Gazebo serves as a virtual representation of the proposed hardware platform.

3.3 Software Interfaces

The Autonomous Warehouse Fleet Management System interacts with multiple software components throughout its operation.

The system interfaces include:

- ***ROS 2 Jazzy*** *for robot communication and middleware services.*
- ***Gazebo Harmonic*** *for warehouse simulation.*
- ***Navigation2 (Nav2)*** *for autonomous robot navigation.*
- ***RViz2*** *for robot visualization and debugging.*
- ***Python 3*** *for application development.*

- ***Streamlit*** *for dashboard development.*
- ***SQLite*** *for storing robot information, delivery records, and system logs.*
- ***Git and GitHub*** *for version control and collaborative development.*

Communication between software modules shall utilize ROS 2 topics, services, and actions.

3.4 Communications Interfaces

Communication within the Autonomous Warehouse Fleet Management System shall be based on the ROS 2 communication framework.

The communication mechanisms include:

- ***ROS 2 Topics*** *for continuous exchange of robot status, sensor readings, and navigation updates.*
- ***ROS 2 Services*** *for request-response operations such as robot registration and task assignment.*
- ***ROS 2 Actions*** *for long-running operations including autonomous delivery execution.*
- ***Wi-Fi*** *(future implementation) for communication between the Fleet Management System and physical robots.*

The communication framework shall ensure reliable message exchange, synchronized fleet coordination, and efficient task execution within the warehouse environment

4. System Features

The Autonomous Warehouse Fleet Management System consists of several functional modules that collectively enable centralized coordination, monitoring, and management of multiple autonomous warehouse robots. Each system feature described below defines the functionality, priority, expected system behavior, and functional requirements necessary for successful operation.

4.1 Robot Registration and Fleet Initialization

4.1.1 Description and Priority

The Robot Registration and Fleet Initialization feature enables the Fleet Management System to register autonomous robots before warehouse operations begin. Each robot is assigned a unique identifier and initialized with its operational status, availability, and communication parameters. This feature establishes communication between the robots and the Fleet Management System and serves as the foundation for all subsequent warehouse operations.

Priority: *High*

4.1.2 Stimulus/Response Sequences

Stimulus:*The warehouse operator launches the Fleet Management System or introduces a new autonomous robot into the warehouse simulation.*

Response:

- *The Fleet Management System verifies the robot.*
- *A unique Robot ID is assigned.*
- *The robot status is initialized as Available.*
- *The robot appears on the Streamlit dashboard.*
- *The Fleet Manager begins monitoring the robot.*

4.1.3 Functional Requirements

FR-1.1 The system shall register autonomous robots before task allocation begins.

FR-1.2 Each robot shall be assigned a unique Robot ID.

FR-1.3 The system shall maintain the operational status of every registered robot.

FR-1.4 The dashboard shall display all registered robots and their current status.

FR-1.5 The system shall reject duplicate robot registrations.

FR-1.6 The Fleet Management System shall verify successful communication with each robot before marking it as available

FR-1.7 The system shall update the dashboard whenever a robot joins or leaves the fleet.

4.2 Delivery Task Management

4.2.1 Description and Priority

The Delivery Task Management feature enables warehouse operators to create, assign, monitor, and manage package delivery requests within the warehouse. The Fleet Management System is responsible for selecting an appropriate robot based on its availability and current operational status. The system continuously tracks task progress from assignment until successful completion.

Priority: High

4.2.2 Stimulus/Response Sequences

Stimulus:\u200b*The warehouse operator creates a new delivery request by selecting a pickup location and a destination through the Streamlit dashboard.*

Response:

- *The system validates the delivery request.*
- *The Fleet Management System searches for an available robot.*
- *The selected robot is assigned the delivery task.*
- *The robot status changes from Available to Busy.*
- *The task status changes to In Progress.*
- *The dashboard updates the assigned robot, delivery route, and task progress in real time.*
- *Upon successful delivery, the task status changes to Completed, and the robot becomes Available again*

4.2.3 Functional Requirements

FR-2.1 The system shall allow warehouse operators to create delivery requests.

FR-2.2 Each delivery request shall contain a unique Task ID.

FR-2.3 Each delivery request shall include a pickup location and destination.

FR-2.4 The Fleet Management System shall assign delivery tasks only to available robots.

FR-2.5 The system shall update the operational status of the assigned robot during task execution.

FR-2.6 The dashboard shall display the current status of every delivery task.

FR-2.7 The system shall record the start time and completion time of each delivery task.

FR-2.8 The system shall mark the task as completed after the robot successfully reaches the destination.

FR-2.9 The system shall automatically return the robot to the Available state after task completion.

FR-2.10 The system shall maintain a history of completed delivery tasks for future analysis

4.3 Fleet Monitoring

4.3.1 Description and Priority

The Fleet Monitoring feature enables the Fleet Management System to continuously monitor the operational status of all registered autonomous robots within the warehouse. It provides real-time visibility into robot activities, task execution, lane occupancy, and communication status through the Streamlit dashboard. This feature ensures that the Fleet Manager always has up-to-date information required for efficient task allocation, traffic coordination, and warehouse supervision

Priority: High

4.3.2 Stimulus/Response Sequences

Stimulus:*A robot changes its operational state during warehouse operation, such as accepting a task, moving to a destination, waiting for lane clearance, completing a delivery, or encountering an obstacle*

Response:

- *The robot transmits its updated status to the Fleet Management System.*
- *The Fleet Management System updates the robot's operational information.*
- *The Streamlit dashboard refreshes the robot's status in real time.*
- *The Fleet Manager verifies whether the robot requires further action.*
- *If an abnormal condition is detected, the system generates an appropriate notification for the warehouse operator*

4.3.3 Functional Requirements

FR-3.1 The system shall continuously monitor the operational status of all registered robots.

FR-3.2 The system shall maintain the current location of every robot within the warehouse simulation.

FR-3.3 The system shall display the current operational state of every robot on the dashboard.

FR-3.4 The system shall monitor the delivery task currently assigned to each robot.

FR-3.5 The system shall detect robots that are in an idle, busy, waiting, or unavailable state.

FR-3.6 The system shall update robot information whenever a status change occurs.

FR-3.7 The system shall notify the warehouse operator whenever a robot encounters an unexpected operational event.

FR-3.8 The dashboard shall provide real-time visualization of the fleet status.

FR-3.9 The system shall maintain historical operational logs for each robot.

FR-3.10 The Fleet Management System shall provide operational information required for traffic coordination and task allocation

4.4 Dynamic Traffic Coordination

4.4.1 Description and Priority

The Dynamic Traffic Coordination feature enables the Fleet Management System to regulate robot movement within the warehouse by coordinating lane usage and preventing traffic conflicts. Whenever multiple robots request access to the same warehouse lane, the Fleet Management System evaluates lane availability and grants access based on predefined traffic coordination rules. This centralized approach minimizes congestion, prevents collisions, and ensures efficient warehouse operations.

Priority: High

4.4.2 Stimulus/Response Sequences

Stimulus:*An autonomous robot requests access to a warehouse lane while another robot is currently occupying or has already reserved the same lane*

Response:

- *The Fleet Management System receives the lane reservation request.*
- *The system checks the current occupancy and reservation status of the requested lane.*
- *If the lane is available, the reservation is approved and assigned to the requesting robot.*
- *If the lane is occupied or reserved, the request is temporarily denied.*
- *The requesting robot enters a waiting state until the lane becomes available.*
- *Once the lane is released, the Fleet Management System grants access to the next eligible robot.*
- *The dashboard updates the lane status and robot states in real time*

4.4.3 Functional Requirements

FR-4.1 The system shall maintain the occupancy status of every warehouse lane.

FR-4.2 The system shall allow robots to request lane access before entering a shared lane.

FR-4.3 The Fleet Management System shall evaluate lane availability before approving a lane request.

FR-4.4 The system shall permit only one robot to reserve or occupy a shared lane at any given time.

FR-4.5 The system shall reject conflicting lane reservation requests.

FR-4.6 The system shall place robots in a Waiting state if the requested lane is unavailable**.**

FR-4.7 The system shall automatically release a lane after the occupying robot exits the reserved area.

FR-4.8 The Fleet Management System shall grant the released lane to the next eligible robot.

FR-4.9 The dashboard shall display the real-time reservation status of every warehouse lane.

FR-4.10 The system shall maintain a log of lane reservations and traffic events.

4.5 Dashboard and Analytics

4.5.1 Description and Priority

The Dashboard and Analytics feature provides warehouse operators with a centralized interface for monitoring and managing fleet operations. The dashboard presents real-time information about robot activities, delivery tasks, lane occupancy, traffic status, and warehouse performance. It also maintains historical records to support operational analysis and decision-making.

Priority: High

4.5.2 Stimulus/Response Sequences

Stimulus:*The warehouse operator accesses the Streamlit dashboard or the Fleet Management System detects a change in the operational status of the warehouse*

Response:

- *The dashboard retrieves the latest fleet information from the Fleet Management System.*
- *Robot locations and operational states are updated in real time.*
- *Active delivery tasks and their progress are displayed.*
- *Lane occupancy and traffic status are refreshed.*
- *Notifications are displayed for completed deliveries, obstacle detections, and traffic conflicts.*
- *Warehouse performance statistics are updated automatically.*

4.5.3 Functional Requirements

FR-5.1 The system shall provide a real-time dashboard for warehouse monitoring.

FR-5.2 The dashboard shall display the operational status of every registered robot**.**

FR-5.3 The dashboard shall display the current location of each robot within the warehouse simulation.

FR-5.4 The dashboard shall display the status of all active delivery tasks.

FR-5.5 The dashboard shall display the occupancy status of warehouse lanes.

FR-5.6 The dashboard shall notify operators about completed deliveries, traffic conflicts, and obstacle-related events.

FR-5.7 The dashboard shall maintain a history of completed delivery tasks.

FR-5.8 The system shall generate operational statistics, including the number of completed deliveries and robot utilization.

FR-5.9 The dashboard shall automatically refresh whenever fleet information changes.

FR-5.10 The system shall provide historical logs for performance analysis and troubleshooting

5. Other Nonfunctional Requirements

5.1 Performance Requirements

The Autonomous Warehouse Fleet Management System shall satisfy the following performance requirements:

- *The system shall update robot status on the dashboard in near real time.*
- *The Fleet Management System shall process delivery requests with minimal delay.*
- *The dashboard shall refresh automatically whenever robot or task information changes.*
- *The system shall support simultaneous monitoring of multiple autonomous robots.*
- *Lane reservation requests shall be processed before robots enter shared warehouse lanes.*
- *The system shall maintain consistent performance throughout continuous simulation*

5.2 Safety Requirements

The system shall ensure safe operation of multiple autonomous robots within the warehouse simulation through centralized traffic coordination.

Safety requirements include:

- *The system shall prevent multiple robots from occupying the same reserved lane simultaneously.*
- *Robots shall wait for lane clearance before entering occupied lanes.*
- *The Fleet Management System shall detect traffic conflicts before granting lane access.*
- *The system shall notify warehouse operators whenever abnormal robot behavior is detected.*
- *The simulation shall prevent conflicting task assignments*

5.3 Security Requirements

Although the current implementation is intended for simulation, the following security requirements shall be considered:

- Only authorized users shall access the Fleet Management Dashboard.
- Robot communication shall occur only through registered ROS2 nodes.
- System logs shall be protected from unauthorized modification.
- Future cloud integration shall utilize secure communication protocols

5.4 Software Quality Attributes

The Autonomous Warehouse Fleet Management System shall satisfy the following software quality attributes:

- ***Reliability:*** *The system shall maintain stable fleet coordination throughout warehouse operations.*
- ***Scalability:*** *The architecture shall support future expansion to additional robots and*

- *warehouse zones.*
- ***Maintainability:*** *The software shall follow a modular architecture to simplify maintenance and future enhancements.*
- ***Usability:*** *The dashboard shall provide an intuitive interface suitable for warehouse operators.*
- ***Availability:*** *The Fleet Management System shall remain operational throughout warehouse simulation.*
- ***Extensibility:*** *The architecture shall support future integration with physical robots, AI, cloud services, and IoT technologies*

5.5 Business Rules

The Autonomous Warehouse Fleet Management System shall operate according to the following business rules:

- *Every delivery request shall be assigned a unique Task ID.*
- *Every autonomous robot shall possess a unique Robot ID.*
- *A robot shall execute only one delivery task at a time.*
- *Only one robot may reserve or occupy a shared warehouse lane at any given time.*
- *A delivery task shall be assigned only to an available robot.*
- *Completed delivery tasks shall be stored for future analysis and reporting*

6. Other Requirements

The following additional requirements support the development and maintenance of the Autonomous Warehouse Fleet Management System:

- *The project shall utilize Git and GitHub for version control.*
- *The software shall follow modular programming principles.*
- *ROS2 packages shall be organized according to recommended ROS2 workspace standards.*
- *The project shall include comprehensive documentation, including the SRS, user manual, and GitHub README.*
- *The software shall be designed to facilitate future integration with physical robots and cloud-based warehouse management systems*

Appendix A: Glossary

Term	**Description**
AWFMS	*Autonomous Warehouse Fleet Management System*
ROS 2	*Robot Operating System 2*
Gazebo	*Robot simulation platform*
Nav2	*ROS2 Navigation Stack*
Fleet Manager	*Centralized software responsible for coordinating all warehouse robots*
Streamlit	*Python framework used to develop the dashboard*
Task ID	*Unique identifier assigned to every delivery request*
Robot ID	*Unique identifier assigned to every robot*

Appendix B: Analysis Models

The following analysis models have been developed to support the design and implementation of the Autonomous Warehouse Fleet Management System:

- *High-Level System Architecture Diagram*
- *Use Case Diagram*
- *Activity Diagram*
- *Sequence Diagram*
- *Class Diagram*
- *State Diagram*
- *Deployment Diagram*

Appendix C: Future Enhancements

Future enhancements planned for the Autonomous Warehouse Fleet Management System include:

- *Integration with physical autonomous robots.*
- *AI-based task allocation and route optimization.*
- *Cloud-based fleet monitoring.*
- *IoT-enabled warehouse infrastructure.*
- *Battery-aware scheduling.*
- *Multi-warehouse fleet management.*
- *Predictive maintenance using machine learning.*
- *Computer vision-based obstacle detection.*
- *MQTT-based communication with distributed robots.*